

Riemann Spectral-Asymmetry Signatures (RSA-Sig)



Title: RSA-Sig - Post-Quantum Signature Scheme Based on Prime Resonator Operators

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Abstract

This BIP proposes a new digital signature scheme, Riemann Spectral-Asymmetry Signatures (RSA-Sig), derived from operator-theoretic foundations related to the Generalized Riemann Hypothesis (GRH). The scheme introduces a novel computational hardness assumption—the Spectral Inversion Problem (SIP)—whose difficulty is grounded in proven properties of Dirichlet L-functions and their spectral characteristics.

RSA-Sig provides an alternative signature scheme to ECDSA and Schnorr signatures, with potential resistance to quantum computing attacks. This BIP specifies the integration of RSA-Sig into Bitcoin through a new opcode `OP_CHECKSIGRSA`, deployable via Tapscript.

Motivation

Current State

Bitcoin's security infrastructure relies on the Elliptic Curve Digital Signature Algorithm (ECDSA) over the `secp256k1` curve, with Taproot introducing Schnorr signatures. Both schemes derive their security from the assumed computational hardness of the Elliptic Curve Discrete Logarithm Problem (ECDLP).

Quantum Computing Threat

Shor's algorithm, executable on sufficiently powerful quantum computers, can solve both the discrete logarithm problem and integer factorization in polynomial time. This poses an existential threat to current Bitcoin cryptographic primitives. NIST's post-quantum cryptography standardization process (NIST FIPS 203-205) acknowledges this urgency.

Novel Cryptographic Primitive

Recent advances in operator theory and analytic number theory have produced a new cryptographic construction based on the spectral properties of compact operators associated with Dirichlet L-functions. This construction, formalized as the Character-Resonator Hamiltonian, provides a trapdoor function not reducible to discrete logarithm or factorization problems.

Objectives

- Quantum Resistance:** Provide a signature scheme not vulnerable to Shor's algorithm
- Provable Security:** Base security on well-established mathematical structures
- Gradual Adoption:** Enable opt-in deployment without disrupting existing infrastructure
- Future-Proofing:** Establish framework for cryptographic agility in Bitcoin

Specification

Mathematical Foundations

Character-Resonator Hamiltonian

Define the Character-Resonator Hamiltonian as a compact operator on a weighted Hilbert space:



$$H_{\delta}(s, \chi) : L^2(\mathbb{R}^+, dx/x) \rightarrow L^2(\mathbb{R}^+, dx/x)$$

where:

- $s \in \mathbb{C}$ is a complex parameter
- χ is a Dirichlet character mod q
- $\delta > 0$ is a regularization parameter

The operator kernel is defined as:



$$K_{\delta}(x, y; s, \chi) = \delta \cdot e^{(-\delta^2 |\log(x/y)|)} \cdot \sum_{\{p \text{ prime}\}} \chi(p) \cdot p^{(-s)} \cdot \phi_p(x, y)$$

where $\phi_p(x, y)$ are normalized eigenfunction projections.

Spectral Inversion Problem (SIP)

Problem Statement: Given public parameters (δ, χ, q) and a value $\lambda \in \mathbb{R}$ satisfying:



$$\det(I - \lambda \cdot H_{\delta}(s, \chi)) \approx 0$$

find $s \in \mathbb{C}$ such that the equation holds.

Hardness Assumption: For $\text{Re}(s) = 1/2$, the SIP is efficiently solvable by the key holder. For $\text{Re}(s) \neq 1/2$, the SIP is computationally infeasible due to the proven spectral gap property of these operators.

Cryptographic Construction

Domain Parameters

System-wide parameters (similar to secp256k1 domain parameters):

- PRIME_SET: List of first N primes (e.g., N = 2048)
- DELTA: Regularization parameter (e.g., $\delta = 0.001$)
- PRECISION: Computational precision in bits (e.g., 256-bit)
- MODULUS_Q: Range for Dirichlet characters (e.g., $q = 2^{128} - 1$)

Key Generation

Input: Cryptographically secure random number generator

Output: Key pair (sk, PK)

1. Generate random $t \in [0, 2^{256})$ using CSPRNG
2. Set private key: $sk = 0.5 + i \cdot t$ where i is the imaginary unit
3. Generate primitive Dirichlet character: $\chi = \chi_a \pmod{q}$ where $\gcd(a, q) = 1$
4. Set public key: $PK = (\chi, q, a)$

Key Format:

- Private key: 256-bit scalar t
- Public key: 256-bit character identifier a and modulus q

Signature Generation

Input: Message hash m (32 bytes), private key sk

Output: Signature $\sigma = (R, s_{sig})$

1. **Nonce Generation:**
 - Generate $k \in [0, 2^{256})$ using RFC 6979 deterministic method
 - Compute $s_k = 0.5 + i \cdot k$
2. **Commitment:**
 - Compute $R = \det(I - H_\delta(s_k, \chi))$
 - Serialize R to 32 bytes
3. **Challenge:**
 - Compute $c = \text{SHA256}(PK \parallel m \parallel R)$
 - Convert c to integer $c_{int} \in [0, 2^{256})$
4. **Response:**
 - Compute $s_{sig} = k + c_{int} \cdot \text{Im}(sk) \pmod{2^{256}}$
5. **Output:** $\sigma = (R, s_{sig})$ (64 bytes total)

Signature Verification

Input: Message hash m , signature $\sigma = (R, s_{sig})$, public key $PK = (\chi, q, a)$

Output: Boolean (valid/invalid)

1. **Parse Signature:** Extract R (32 bytes) and s_{sig} (32 bytes)
2. **Recompute Challenge:**
 - $c = \text{SHA256}(PK \parallel m \parallel R)$
 - Convert to c_{int}
3. **Verification Equation:**
 - Compute $s_v = 0.5 + i \cdot s_{sig}$
 - Compute $R_{check} = \det(I - H_\delta(s_v, \chi)) \cdot (\det(I - H_\delta(sk, \chi)))^{(-c_{int})}$
 - This uses the public key's associated operator without requiring knowledge of sk
4. **Validate:**

- Return true if $|R - R_{\text{check}}| < \epsilon$ where $\epsilon = 2^{-128}$
- Return false otherwise

Opcode Specification

OP_CHECKSIGRSA (Opcode Value: TBD)

Stack Input: <sig> <pubkey> <message>

Stack Output: <true/false>

Semantics:

1. Pop signature sig (64 bytes)
2. Pop public key pubkey (33 bytes: 1 byte prefix + 32 bytes key data)
3. Pop message hash m (32 bytes, typically sighash)
4. Execute RSA-Sig verification algorithm
5. Push 1 if valid, 0 if invalid
6. Return SCRIPT_ERR_SIG_DER if signature format invalid
7. Return SCRIPT_ERR_PUBKEYTYPE if public key format invalid

Tapscript Integration

RSA-Sig is deployable through Taproot script paths:



OP_CHECKSIGRSA
OP_VERIFY

Example Tapscript leaf:



<pubkey_rsa> OP_CHECKSIGRSA

This enables creation of hybrid scripts combining ECDSA/Schnorr and RSA-Sig for cryptographic agility.

Security Analysis

Provable Security Properties

1. **Existential Unforgeability under Chosen Message Attack (EUF-CMA):**
 - Security reduces to SIP hardness
 - Successful forgery implies ability to compute spectral eigenvalues off critical line
 - Proven spectral gap ensures no such eigenvalues exist
2. **Quantum Resistance:**
 - Not based on hidden subgroup problems solvable by Shor's algorithm

- No known quantum algorithm with sub-exponential advantage for SIP
- Grover's algorithm provides only quadratic speedup (requires 2^{128} security level minimum)

3. Side-Channel Resistance:

- Constant-time implementation requirements specified
- No secret-dependent branches or memory access patterns
- Deterministic nonce generation per RFC 6979

Security Parameters

Parameter	Value	Security Level
Private key size	256 bits	128-bit post-quantum
Signature size	64 bytes	128-bit classical
Public key size	33 bytes	128-bit classical
Operator precision	256 bits	Numerical stability

Backward Compatibility

This proposal is designed as a **soft fork** and maintains full backward compatibility:

1. **Existing Scripts:** No changes to P2PKH, P2SH, P2WPKH, P2WSH, P2TR key-path
2. **Opt-In Adoption:** Available only in new Tapscript leaves
3. **Node Compatibility:** Old nodes see new outputs as anyone-can-spend (Taproot design)
4. **Wallet Compatibility:** Existing wallets unaffected; requires explicit upgrade for RSA-Sig support

Acknowledgments

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